

AAYUSH MISHRA

Software Engineer | Python, FastAPI, REST APIs | AI-native Applications

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PROFESSIONAL SUMMARY

Software Engineer with strong foundations in **Data Structures & Algorithms, Object-Oriented Programming, and AI/ML**. Experienced in developing production-grade backend applications, **REST APIs**, AI systems, and AI-enabled automation using Java, Python, SQL, FastAPI, Spring Boot, PostgreSQL, Redis, and **LLM APIs**. Proficient in API integration, authentication, multi-tenant systems, caching, automated testing, CI/CD, and building reliable, secured software solutions.

WORK EXPERIENCE

Software Engineer - Technology & AI

Jul 2026 - Present

Memoire Ventures Pvt. Ltd.

Bangalore, India

- Developed an **AI-powered Google Review automation workflow** using n8n, Python, and LLM APIs to generate and publish review responses, incorporating an automated issue-flagging alert system.
- Built **CafeOS**, a multi-tenant restaurant operations platform; implemented JWT-based authentication and tenant setup, currently developing the SOP & checklist management module.

Technologies / Skills Used : Java, Spring Boot, Supabase, PostgreSQL, Redis, JWT Auth, n8n, Python

PROJECTS

AI-Powered Research Assistant CLI | 🐙 Github

Jul 2026 - Aug 2026

Individual Project

- Built an end-to-end **RAG system** in Python, implementing document ingestion, **200-token chunking with overlap**, **embedding generation**, **cosine-similarity** retrieval, and **LLM synthesis**, achieving **80% source-level Recall@5** across **20 curated queries**.
- Implemented a **persistent semantic retrieval** engine using SQLite and NumPy, indexing **6,760+ document chunks** across **74 topics** and supporting provider-specific embeddings.
- Engineered **modular AI infrastructure** with interchangeable Cohere/Gemini embedding providers, persistent caching, and automated testing, achieving **56 passing tests** and **96% core-module coverage**.

Tech Stack : Python, SQLite, RAG, GitHub Actions, Claude API SDK, Cohere Embed API

URL Shortener with Analytics | 🐙 Github

Apr 2026 - Jun 2026

Individual Project

- Developed a **RESTful URL shortening service** using an indexed **PostgreSQL schema** with a layered Controller–Service–Repository architecture and integrated **click-tracking analytics**.
- Built REST APIs for **URL generation, redirection, and data retrieval**, using Redis caching to reduce read latency by **~60%**.

Tech Stack : Java, Spring Boot, PostgreSQL, Redis, Docker

Distributed API Rate Limiter Service | 🐙 Github

Jan 2026 - Apr 2026

Individual Project

- Engineered a **Redis-backed rate limiting service** utilizing Fixed Window, Sliding Window, and Token Bucket algorithms through atomic Redis operations (INCR, EXPIRE, Lua scripting).
- Developed configurable **REST APIs with HTTP 429 enforcement** and TTL-based distributed state management.
- Conducted load testing to approximately **2000 requests/sec** using wrk with 100 concurrent connections, and deployed via **Docker**.

Tech Stack : Java, Spring Boot, Redis, Docker

SKILLS

Programming Languages : Python, Java, SQL, C++, TypeScript

Frameworks & Libraries : FastAPI, Pydantic, React, Spring Boot

Databases : PostgreSQL, MySQL, Redis, SQLite, MongoDB, ChromaDB

Tools & Platforms : Docker, AWS, CI/CD, Git, GitHub, Linux, GitHub Actions, Postman, Hugging Face

AI/ML : LLM, RAG, AI Agents, Vector DB, Embeddings, PyTorch, TensorFlow, LangChain, Fine-Tuning

Testing : PyTest, JUnit, Mockito

EDUCATIONS

RV Institute of Technology and Management | Bangalore

2022 - 2026

B.E. in Electronics & Communication Engineering

CGPA : 7.8/10

Relevant Coursework

Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design, REST APIs

ACHIEVEMENTS

Robofiesta Hack-A-Day Winner

Nov 2025

Robotics Club, RVITM

Secured 1st place among 18 teams at the 1-day hackathon, for developing a face-recognition attendance system with ~94% accuracy, tested on 60-70 individuals.